

## Config File Demo

The virtual installation loads from a configuration file describing the start states of each element. The configuration file is JSON. The following is an example of a configuration file.

---

```
{
  "coordinateSystems": [
    {
      "name": "world",
      "type": "coordinateSystem",
      "description": "Global coordinate system for the entire setup.",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    },
    {
      "name": "armature",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "world",
      "description": "Coordinate system for the armature unit, based on world coordinates.",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    },
    {
      "name": "plinth",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "world",
      "description": "Coordinate system for the plinth, based on world coordinates.",
      "location": {"x": 0, "y": 0, "z": 2.0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    },
    {
      "name": "female_1_base",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "armature",
      "description": "Base coordinate system for female 1, based on armature coordinates.",
      "location": {"x": 0, "y": 68, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    },
    {
      "name": "female_2_base",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "armature",
      "description": "Base coordinate system for female 2, based on armature coordinates.",
      "location": {"x": 58.5, "y": -34, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 240}
    },
    {
      "name": "female_3_base",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "armature",
      "description": "Base coordinate system for female 3, based on armature coordinates.",
      "location": {"x": -58.5, "y": -34, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 120}
    },
    {
      "name": "beam",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "armature",
      "description": "Coordinate system for the beam unit, based on armature coordinates.",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    },
    {
      "name": "beam_horizontal_control",
      "type": "coordinateSystem",
      "parentCoordinateSystem": "beam",
      "description": "Coordinate system for the horizontal control subsystem of the beam."
    }
  ]
}
```

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```
"location": {"x": 0, "y": 0, "z": 0},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0},
"rangeOfMotion": {
  "yaw": {"min": -180, "max": 180}
}
},
{
  "name": "male_1_base",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "beam",
  "description": "Base coordinate system for male 1, based on beam coordinates.",
  "location": {"x": 0, "y": -23.5, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
  "name": "male_2_base",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "beam",
  "description": "Base coordinate system for male 2, based on beam coordinates.",
  "location": {"x": 0, "y": -23.5, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 180}
},
{
  "name": "female_1_horizontal_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_1_base",
  "description": "Coordinate system for the horizontal control subsystem of female 1.",
  "location": {"x": 0, "y": 0, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "yaw": {"min": -30, "max": 30}
  }
},
{
  "name": "female_1_vertical_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_1_horizontal_control",
  "description": "Coordinate system for the vertical control subsystem of female 1.",
  "location": {"x": 0, "y": -9.872, "z": -40.5},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "roll": {"min": -15, "max": 15}
  }
},
{
  "name": "female_2_horizontal_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_2_base",
  "description": "Coordinate system for the horizontal control subsystem of female 2.",
  "location": {"x": 0, "y": 0, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "yaw": {"min": -30, "max": 30}
  }
},
{
  "name": "female_2_vertical_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_2_horizontal_control",
  "description": "Coordinate system for the vertical control subsystem of female 2.",
  "location": {"x": 0, "y": -9.872, "z": -40.5},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "roll": {"min": -15, "max": 15}
  }
},
{
  "name": "female_3_horizontal_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_3_base",
  "description": "Coordinate system for the horizontal control subsystem of female 3.",
```

## Config File Demo

```

"location": {"x": 0, "y": 0, "z": 0},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0},
"rangeOfMotion": {
  "yaw": {"min": -30, "max": 30}
}
},
{
  "name": "female_3_vertical_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "female_3_horizontal_control",
  "description": "Coordinate system for the vertical control subsystem of female 3.",
  "location": {"x": 0, "y": -9.872, "z": -40.5},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "roll": {"min": -15, "max": 15}
  }
},
{
  "name": "male_1_horizontal_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "male_1_base",
  "description": "Coordinate system for the horizontal control subsystem of male 1.",
  "location": {"x": 0, "y": 0, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "yaw": {"min": -60, "max": 60}
  }
},
{
  "name": "male_2_horizontal_control",
  "type": "coordinateSystem",
  "parentCoordinateSystem": "male_2_base",
  "description": "Coordinate system for the horizontal control subsystem of male 2.",
  "location": {"x": 0, "y": 0, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "yaw": {"min": -60, "max": 60}
  }
}
],
"units": [
  {
    "name": "armature",
    "type": "location",
    "thisCoordinateSystem": "world",
    "location": {"x": 0, "y": 0, "z": 0},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "childElements": [
      {
        "name": "plinth",
        "type": "location",
        "thisCoordinateSystem": "armature",
        "parentCoordinateSystem": "world",
        "location": {"x": 0, "y": 0, "z": 2},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
      }
    ],
    {
      "name": "female 1",
      "type": "location",
      "thisCoordinateSystem": "armature",
      "parentCoordinateSystem": "world",
      "location": {"x": 0, "y": 68, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
      "childElements": [
        {
          "name": "female",
          "type": "origin",
          "thisCoordinateSystem": "female_1_base",
          "parentCoordinateSystem": "armature",
          "location": {"x": 0, "y": 0, "z": 0},
          "orientation": {"roll": 0, "pitch": 0, "yaw": 0},

```

# Config File Demo

```

"childElements": [
  {
    "name": "horizontal control subsystem",
    "type": "actuator",
    "thisCoordinateSystem": "female_1_horizontal_control",
    "parentCoordinateSystem": "female_1_base",
    "location": {"x": 0, "y": 0, "z": 0},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "rangeOfMotion": {
      "yaw": {"min": -30, "max": 30}
    },
    "childElements": [
      {
        "name": "female speaker",
        "type": "actuator",
        "thisCoordinateSystem": "female_1_horizontal_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": -2, "y": -0.25, "z": -34.75},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
      },
      {
        "name": "female microphone",
        "type": "sensor",
        "thisCoordinateSystem": "female_1_horizontal_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": 2, "y": -0.25, "z": -34.75},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
      },
      {
        "name": "vertical control subsystem",
        "type": "actuator",
        "thisCoordinateSystem": "female_1_vertical_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": 0, "y": -9.872, "z": -40.5},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
        "rangeOfMotion": {
          "roll": {"min": -15, "max": 15}
        },
        "childElements": [
          {
            "name": "female light sensor",
            "type": "sensor",
            "thisCoordinateSystem": "female_1_vertical_control",
            "parentCoordinateSystem": "vertical control subsystem",
            "location": {"x": 0, "y": 0, "z": 0},
            "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
          }
        ]
      }
    ]
  }
]
},
{
  "name": "female 2",
  "type": "location",
  "thisCoordinateSystem": "armature",
  "parentCoordinateSystem": "world",
  "location": {"x": 58.5, "y": -34, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "childElements": [
    {
      "name": "female",
      "type": "origin",
      "thisCoordinateSystem": "female_2_base",
      "parentCoordinateSystem": "armature",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
      "childElements": [

```

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```
{
  "name": "horizontal control subsystem",
  "type": "actuator",
  "thisCoordinateSystem": "female_2_horizontal_control",
  "parentCoordinateSystem": "female_2_base",
  "location": {"x": 0, "y": 0, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "rangeOfMotion": {
    "yaw": {"min": -30, "max": 30}
  },
},
"childElements": [
  {
    "name": "female speaker",
    "type": "actuator",
    "thisCoordinateSystem": "female_2_horizontal_control",
    "parentCoordinateSystem": "horizontal control subsystem",
    "location": {"x": -2, "y": -0.25, "z": -34.75},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
  },
  {
    "name": "female microphone",
    "type": "sensor",
    "thisCoordinateSystem": "female_2_horizontal_control",
    "parentCoordinateSystem": "horizontal control subsystem",
    "location": {"x": 2, "y": -0.25, "z": -34.75},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
  },
  {
    "name": "vertical control subsystem",
    "type": "actuator",
    "thisCoordinateSystem": "female_2_vertical_control",
    "parentCoordinateSystem": "horizontal control subsystem",
    "location": {"x": 0, "y": -9.872, "z": -40.5},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "rangeOfMotion": {
      "roll": {"min": -15, "max": 15}
    },
  },
  "childElements": [
    {
      "name": "female light sensor",
      "type": "sensor",
      "thisCoordinateSystem": "female_2_vertical_control",
      "parentCoordinateSystem": "vertical control subsystem",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    }
  ]
}
]
}
}
},
{
  "name": "female 3",
  "type": "location",
  "thisCoordinateSystem": "armature",
  "parentCoordinateSystem": "world",
  "location": {"x": -58.5, "y": -34, "z": 0},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
  "childElements": [
    {
      "name": "female",
      "type": "origin",
      "thisCoordinateSystem": "female_3_base",
      "parentCoordinateSystem": "armature",
      "location": {"x": 0, "y": 0, "z": 0},
      "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
      "childElements": [
        {

```

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```

    "name": "horizontal control subsystem",
    "type": "actuator",
    "thisCoordinateSystem": "female_3_horizontal_control",
    "parentCoordinateSystem": "female_3_base",
    "location": {"x": 0, "y": 0, "z": 0},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "rangeOfMotion": {
      "yaw": {"min": -30, "max": 30}
    },
    "childElements": [
      {
        "name": "female speaker",
        "type": "actuator",
        "thisCoordinateSystem": "female_3_horizontal_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": -2, "y": -0.25, "z": -34.75},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
      },
      {
        "name": "female microphone",
        "type": "sensor",
        "thisCoordinateSystem": "female_3_horizontal_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": 2, "y": -0.25, "z": -34.75},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
      },
      {
        "name": "vertical control subsystem",
        "type": "actuator",
        "thisCoordinateSystem": "female_3_vertical_control",
        "parentCoordinateSystem": "horizontal control subsystem",
        "location": {"x": 0, "y": -9.872, "z": -40.5},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
        "rangeOfMotion": {
          "roll": {"min": -15, "max": 15}
        },
        "childElements": [
          {
            "name": "female light sensor",
            "type": "sensor",
            "thisCoordinateSystem": "female_3_vertical_control",
            "parentCoordinateSystem": "vertical control subsystem",
            "location": {"x": 0, "y": 0, "z": 0},
            "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
          }
        ]
      }
    ]
  },
  {
    "name": "beam",
    "type": "origin",
    "thisCoordinateSystem": "armature",
    "parentCoordinateSystem": "world",
    "location": {"x": 0, "y": 0, "z": 0},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "childElements": [
      {
        "name": "male 1",
        "type": "location",
        "thisCoordinateSystem": "beam",
        "parentCoordinateSystem": "armature",
        "location": {"x": 0, "y": 23.5, "z": 0},
        "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
        "childElements": [
          {
            "name": "male",

```

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```

"type": "origin",
"thisCoordinateSystem": "male_1_base",
"parentCoordinateSystem": "beam",
"location": {"x": 0, "y": 0, "z": 0},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0},
"childElements": [
{
"name": "horizontal control subsystem",
"type": "actuator",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "male_1_base",
"location": {"x": 0, "y": 0, "z": 0},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0},
"rangeOfMotion": {
"yaw": {"min": -60, "max": 60}
},
"childElements": [
{
"name": "male speaker",
"type": "actuator",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": 0.125, "y": 2.125, "z": -33.75},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male microphone",
"type": "sensor",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": -2.125, "y": 2.6575, "z": -36.5},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male light source",
"type": "actuator",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": 0.125, "y": 2.5, "z": -40.5},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male upper light sensor A",
"type": "sensor",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": 3.13082, "y": -13.65211, "z": -9.82129},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male upper light sensor B",
"type": "sensor",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": -3.13012, "y": -13.96944, "z": -9.82152},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male lower light sensor A",
"type": "sensor",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": -2.88281, "y": 1.89948, "z": -51.40671},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
"name": "male lower light sensor B",
"type": "sensor",
"thisCoordinateSystem": "male_1_horizontal_control",
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": 3.13466, "y": 1.89964, "z": -51.40666},

```

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```

        "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
    }
}
}
}
},
{
    "name": "male 2",
    "type": "location",
    "thisCoordinateSystem": "beam",
    "parentCoordinateSystem": "armature",
    "location": {"x": 0, "y": -23.5, "z": 0},
    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
    "childElements": [
        {
            "name": "male",
            "type": "origin",
            "thisCoordinateSystem": "male_2_base",
            "parentCoordinateSystem": "beam",
            "location": {"x": 0, "y": 0, "z": 0},
            "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
            "childElements": [
                {
                    "name": "horizontal control subsystem",
                    "type": "actuator",
                    "thisCoordinateSystem": "male_2_horizontal_control",
                    "parentCoordinateSystem": "male_2_base",
                    "location": {"x": 0, "y": 0, "z": 0},
                    "orientation": {"roll": 0, "pitch": 0, "yaw": 0},
                    "rangeOfMotion": {
                        "yaw": {"min": -60, "max": 60}
                    }
                },
                {
                    "name": "male speaker",
                    "type": "actuator",
                    "thisCoordinateSystem": "male_2_horizontal_control",
                    "parentCoordinateSystem": "horizontal control subsystem",
                    "location": {"x": 0.125, "y": 2.125, "z": -33.75},
                    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
                },
                {
                    "name": "male microphone",
                    "type": "sensor",
                    "thisCoordinateSystem": "male_2_horizontal_control",
                    "parentCoordinateSystem": "horizontal control subsystem",
                    "location": {"x": -2.125, "y": 2.6575, "z": -36.5},
                    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
                },
                {
                    "name": "male light source",
                    "type": "actuator",
                    "thisCoordinateSystem": "male_2_horizontal_control",
                    "parentCoordinateSystem": "horizontal control subsystem",
                    "location": {"x": 0.125, "y": 2.5, "z": -40.5},
                    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
                },
                {
                    "name": "male upper light sensor A",
                    "type": "sensor",
                    "thisCoordinateSystem": "male_2_horizontal_control",
                    "parentCoordinateSystem": "horizontal control subsystem",
                    "location": {"x": 3.13082, "y": -13.65211, "z": -9.82129},
                    "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
                },
                {
                    "name": "male upper light sensor B",
                    "type": "sensor",
                    "thisCoordinateSystem": "male_2_horizontal_control",

```



## Config File Demo

```
"parentCoordinateSystem": "horizontal control subsystem",
"location": {"x": -3.13012, "y": -13.96944, "z": -9.82152},
"orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
  "name": "male lower light sensor A",
  "type": "sensor",
  "thisCoordinateSystem": "male_2_horizontal_control",
  "parentCoordinateSystem": "horizontal control subsystem",
  "location": {"x": -2.88281, "y": 1.89948, "z": -51.40671},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
},
{
  "name": "male lower light sensor B",
  "type": "sensor",
  "thisCoordinateSystem": "male_2_horizontal_control",
  "parentCoordinateSystem": "horizontal control subsystem",
  "location": {"x": 3.13466, "y": 1.89964, "z": -51.40666},
  "orientation": {"roll": 0, "pitch": 0, "yaw": 0}
}
]
}
}
}
}
}
}
}
}
}
}
```